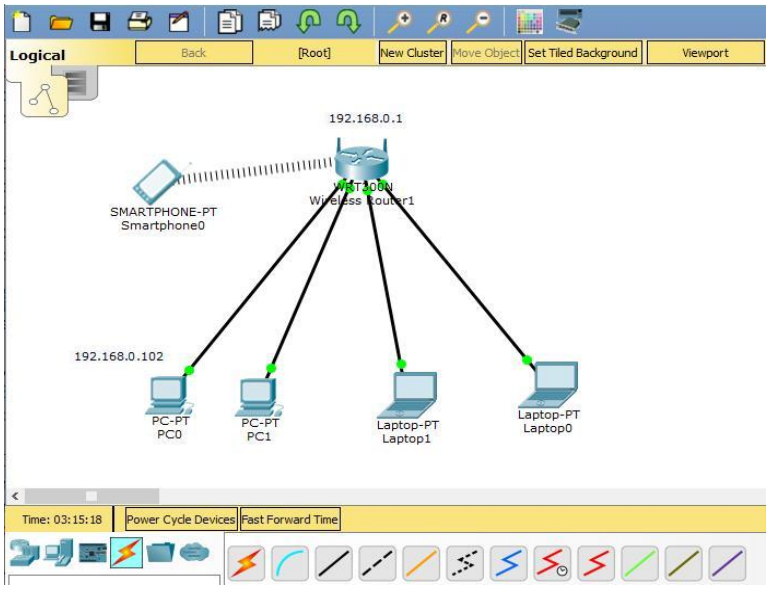
Review and PracticePractical

1.Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.

ANS:

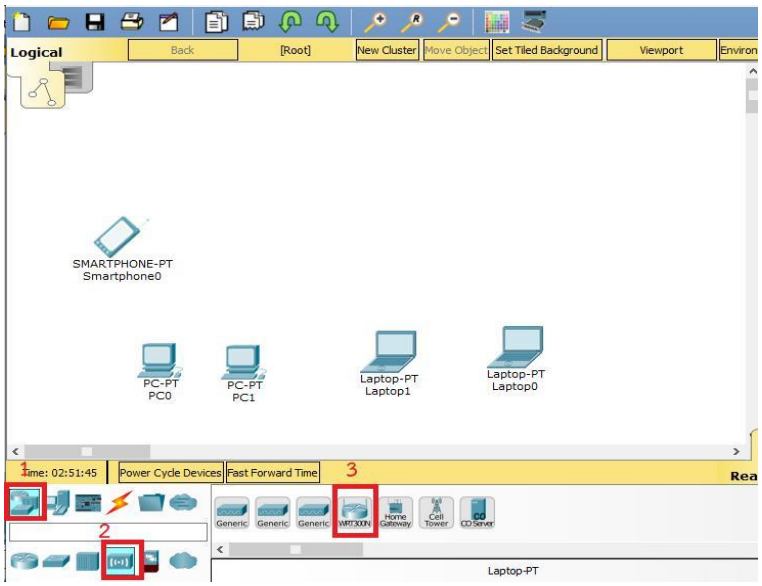
I can guide you to create HomeWiFiNetwork.pkt in Cisco Packet Tracer. I can't directly control the Packet Tracer application on your computer from this chat, so follow these steps.

Step-by-step instructions



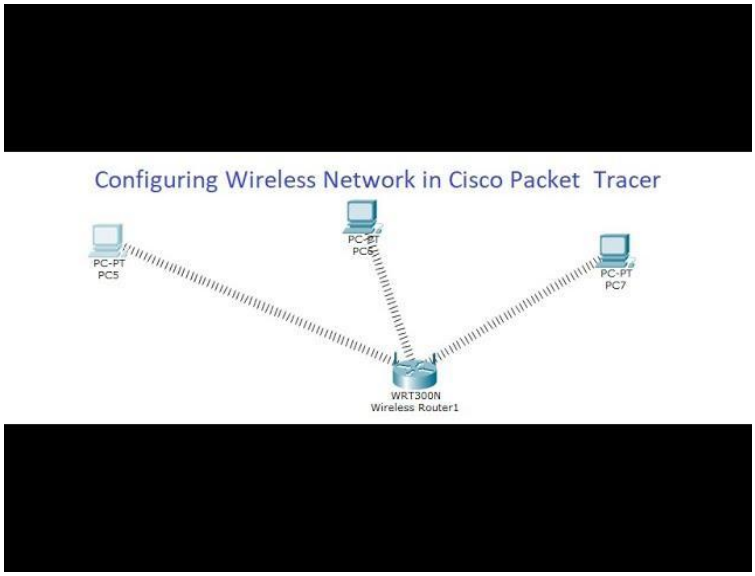
1. Open Cisco Packet Tracer

Launch Packet Tracer and select File → New.



2. Add three PCs

Click End Devices at the bottom, then drag three PCs into the workspace. Rename them PC0, PC1, and PC2.



3. Add a wireless router

Open Network Devices → Wireless Devices and drag a wireless router, such as the WRT300N, into the workspace.

4. Connect the devices

Use Connections (lightning bolt) → Copper Straight-Through to connect each PC's FastEthernet port to a separate LAN Ethernet port on the wireless router. If your assignment specifically requires Wi-Fi connections, use wireless-capable end devices instead.

5. Save the project

Select File → Save As, enter HomeWiFiNetwork.pkt, choose your assignment folder, and click Save.

Expected network diagram

Wireless Router

PC0

PC1

PC2

Three PCs connected to the router's LAN ports using Ethernet cables.

Final check: Make sure all three links turn green, then save the .pkt file.

2.Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

ANS:

Assign Static IP Addresses in Cisco Packet Tracer

Use the following settings for your HomeWiFiNetwork.pkt project. All three PCs will be on the same subnet.

1. IP address table

Device	IP Address	Subnet Mask	Default Gateway
Router (LAN)	192.168.10.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1

2. Configure each PC

1. Click PC0 in Packet Tracer.
2. Open Desktop → IP Configuration.
3. Select Static.
4. Enter the IP address, subnet mask, and default gateway from the table above.
5. Repeat the same steps for PC1 and PC2 using their respective addresses.

3. Configure the router

1. Click the wireless router and open its GUI or Config tab.
2. Find the LAN or local network settings.
3. Set the router's LAN IP address to 192.168.10.1.
4. Set the subnet mask to 255.255.255.0.
5. Save or apply the settings.

Make sure the router's LAN address matches the default gateway entered on each PC. If your router uses a different default LAN address, change it to the address above.

4. Test connectivity

On PC0, open Desktop → Command Prompt and enter:

ping 192.168.10.3

ping 192.168.10.4

ping 192.168.10.1

Expected result: Successful replies indicate that PC0 can communicate with PC1, PC2, and the router. The first ping may time out while ARP resolves addresses; try again if needed.

Finally, save your project using File → Save.

**3.Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router.

Hint: Use the Command Prompt tool on each PC and type 'ping [other PC's IP]'.**

ANS:

Test Connectivity Using the Ping Command in Cisco Packet Tracer

Objective: Verify that all three PCs can communicate with each other and the wireless router.

Step 1: Open Command Prompt

1. Click PC0 in Packet Tracer.
2. Select Desktop → Command Prompt.
3. Type the following commands and press Enter after each command.

ping 192.168.10.1

ping 192.168.10.3

ping 192.168.10.4

Step 2: Test from each PC

Testing Device	Ping Destination	Expected Result
PC0 (192.168.10.2)	Router (192.168.10.1)	Successful
PC0 (192.168.10.2)	PC1 (192.168.10.3)	Successful
PC0 (192.168.10.2)	PC2 (192.168.10.4)	Successful
PC1 (192.168.10.3)	PC2 (192.168.10.4)	Successful
PC1 (192.168.10.3)	Router (192.168.10.1)	Successful
PC2 (192.168.10.4)	Router (192.168.10.1)	Successful

Repeat the relevant commands from PC1 and PC2 to verify connectivity in both directions.

Step 3: Check the results

A successful ping displays a message similar to:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128

- Reply received: The devices can communicate.
- Request timed out: Check the IP addresses, subnet masks, cables, and router LAN settings. Try the ping again, as the first attempt may fail during ARP resolution.

Result

If all the pings succeed, the three PCs and the wireless router are communicating correctly on the 192.168.10.0/24 subnet.

4.Document each step you took to build and configure your Packet Tracer network, including screenshots of your topology and the IP configuration windows for each PC, in a Word or PDF file.

ANS:

HOME Wi-Fi NETWORK
PACKET TRACER REPORT

File name: HomeWiFiNetwork.pkt

1. Aim

To create a simple home network in Cisco Packet Tracer using one wireless router and three PCs, assign static IPv4 addresses, and test connectivity using the ping command.

2. Network Topology

Three PCs are connected to the LAN ports of the wireless router using Ethernet cables. All devices use the 192.168.10.0/24 subnet.

[INSERT SCREENSHOT 1 HERE: Full Packet Tracer topology showing the wireless router and PC0, PC1, and PC2.]
How to capture: In Packet Tracer, arrange the topology so all devices and links are visible, then press Win + Shift + S and capture the workspace.

3. IP Address Plan

Device	IPv4 Address	Subnet Mask	Default Gateway
Router (LAN)	192.168.10.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
PC2	192.168.10.4	255.255.255.0	192.168.10.1

4. Configuration Steps

Step 1 – Create the project: Open Cisco Packet Tracer and select File → New. Save the project as HomeWiFiNetwork.pkt.

Step 2 – Add devices: Add three PCs from End Devices and one wireless router from Network Devices / Wireless Devices. Arrange them in the workspace.

Step 3 – Connect the PCs: Choose Connections (lightning bolt) → Copper Straight-Through. Connect each PC FastEthernet port to a separate LAN Ethernet port on the wireless router.

Step 4 – Configure the router: Set the router's LAN/local IP address to 192.168.10.1 and subnet mask to 255.255.255.0. Apply/save the settings. Menu names can vary by router model.

Step 5 – Configure PC0: Click PC0 → Desktop → IP Configuration → Static. Enter IP address 192.168.10.2, subnet mask 255.255.255.0, and default gateway 192.168.10.1.

Step 6 – Configure PC1: Click PC1 → Desktop → IP Configuration → Static. Enter IP address 192.168.10.3, subnet mask 255.255.255.0, and default gateway 192.168.10.1.

Step 7 – Configure PC2: Click PC2 → Desktop → IP Configuration → Static. Enter IP address 192.168.10.4, subnet mask 255.255.255.0, and default gateway 192.168.10.1.

Step 8 – Save: Select File → Save to save the updated Packet Tracer project.

5. Screenshots of PC IP Configuration

PC0 – IP Configuration

[INSERT SCREENSHOT HERE: PC0 → Desktop → IP Configuration, showing Static selected, IPv4 address 192.168.10.2, subnet mask 255.255.255.0, and default gateway 192.168.10.1.]

Capture tip: Keep the IP Configuration window open and press Win + Shift + S to capture it. Paste the screenshot in this space.

PC1 – IP Configuration

[INSERT SCREENSHOT HERE: PC1 → Desktop → IP Configuration, showing Static selected, IPv4 address 192.168.10.3, subnet mask 255.255.255.0, and default gateway 192.168.10.1.]

Capture tip: Keep the IP Configuration window open and press Win + Shift + S to capture it. Paste the screenshot in this space.

PC2 – IP Configuration

[INSERT SCREENSHOT HERE: PC2 → Desktop → IP Configuration, showing Static selected, IPv4 address 192.168.10.4, subnet mask 255.255.255.0, and default gateway 192.168.10.1.]

Capture tip: Keep the IP Configuration window open and press Win + Shift + S to capture it. Paste the screenshot in this space.

6. Connectivity Test Using Ping

On each PC, open Desktop → Command Prompt and run the following tests.

PC0: ping 192.168.10.1

PC0: ping 192.168.10.3

PC0: ping 192.168.10.4

PC1: ping 192.168.10.1

PC1: ping 192.168.10.4

PC2: ping 192.168.10.1

PC2: ping 192.168.10.2

[OPTIONAL INSERT SCREENSHOT: Command Prompt showing successful ping replies. Record your actual results; do not report success unless the tests worked in Packet Tracer.]

7. Result

The home network is configured with three PCs and one wireless router. The PCs use static IP addresses in the 192.168.10.0/24 subnet and the router's LAN address, 192.168.10.1, as their default gateway. Connectivity is confirmed if the ping tests return successful replies.

8. Submission Checklist

☐ HomeWiFiNetwork.pkt saved

☐ Topology screenshot inserted

☐ PC0 IP Configuration screenshot inserted

☐ PC1 IP Configuration screenshot inserted

☐ PC2 IP Configuration screenshot inserted

☐ Actual ping test results recorded